

Chapter 9: Settings

9.1 Introduction

This chapter gives detailed descriptions of the settings that control the way that your machine works.

9.1.1 List of Settings

Inside the **SETTINGS** tab, the settings are organized into groups. Use the **[UP]** and **[DOWN]** cursor arrow keys to highlight a setting group. Press the **[RIGHT]** cursor arrow key to see the settings in a group, . Press the **[LEFT]** cursor arrow key to return to the setting group list.

To quickly access a single setting, make sure the **SETTINGS** tab is active, type the setting number, and then press **[F1]** or, if a setting is highlighted, press the **[DOWN]** cursor.

Some settings have numerical values that fit in a given range. To change the value of these settings, type the new value and press **[ENTER]**. Other settings have specific available values that you select from a list. For these settings, use the **[RIGHT]** cursor to display the choices. Press **[UP]** and **[DOWN]** to scroll through the options. Press **[ENTER]** to select the option.

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1 - Auto Power Off Timer

This setting is used to automatically power-down the machine after a period of idle time. The value entered in this setting is the number of minutes the machine remains idle until it is powered down. The machine does not power down while a program is running, and the time (number of minutes) starts back at zero anytime a button is pressed or the **[HANDLE JOG]** control is used. The auto-off sequence gives the operator a 15-second warning before power down, at which time pressing any button stops the power down.

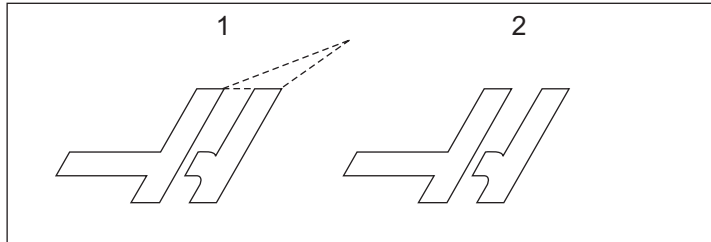
2 - Power Off at M30

If this setting is set to **ON**, the machine powers down at the end of a program (M30). The machine gives the operator a 15-second warning once an M30 is reached. Press any key to interrupt the power-off sequence.

4 - Graphics Rapid Path

This setting changes the way a program is viewed in the Graphics mode. When it is **OFF**, rapid, non-cutting tool motions do not leave a path. When it is **ON**, rapid tool motions leave a dashed line on the screen.

F9.1: Setting 4 -Graphics Rapid Path:[1] All Rapid Tool Motions Shown with a Dashed Line When **ON**. [2] Only Cut Lines Shown When **OFF**.



5 - Graphics Drill Point

This setting changes the way a program is viewed in Graphics mode. When it is **ON**, canned cycle drill locations leave a circle mark on the screen. When it is **OFF**, no additional marks are shown on the graphics display.

6 - Front Panel Lock

When set to **ON**, this Setting disables the Spindle **[FWD]/[REV]** keys and **[TURRET FWD]/[TURRET REV]** keys.

8 - Prog Memory Lock

This setting locks out the memory editing functions (**[ALTER]**, **[INSERT]**, etc.) when it is set to **ON**. This also locks out MDI. Editing functions are not restricted by this setting.

9 - Dimensioning

This setting selects between inch and metric mode. When it is set to **INCH**, the programmed units for X and Z are inches, to 0.0001". When it is set to **MM**, programmed units are millimeters, to 0.001 mm. All offset values are converted when this setting is changed from inches to millimeters, or vice versa. However, changing this setting does not automatically translate a program stored in memory; you must change the programmed axis values for the new units.

When set to **INCH**, the default G code is G20, when set to **MM**, the default G code is G21.

	Inch	Metric
Feed	in/min and in/rev	mm/min and mm/rev
Max Travel	Varies by axis and model	
Minimum programmable dimension	.0001	.001

Axis jog key	Inch	Metric
.0001	.0001 in/jog click	.001 mm/jog click
.001	.001 in/jog click	.01 mm/jog click
.01	.01 in/jog click	.1 mm/jog click
.1	.1 in/jog click	1 mm/jog click

10 - Limit Rapid at 50%

Turning this setting **ON** limits the machine to 50% of its fastest non-cutting axis motion (rapids). This means, if the machine can position the axes at 700 inches per minute (ipm), it is limited to 350 ipm when this setting is **ON**. The control displays a 50% rapid override message, when this setting is **ON**. When it is **OFF**, the highest rapid speed of 100% is available.

17 - Opt Stop Lock Out

The Optional Stop feature is not available when this setting is **ON**.

18 - Block Delete Lock Out

The Block Delete feature is not available when this setting is **ON**.

19 - Feedrate Override Lock

The feedrate override buttons are disabled when this setting is turned **ON**.

20 - Spindle Override Lock

The spindle speed override keys are disabled when this setting is turned **ON**.

21 - Rapid Override Lock

The axis rapid override keys are disabled when this setting is turned **ON**.

22 - Can Cycle Delta Z

This setting specifies the distance to retract the Z Axis to clear chips during a *G73* irregular path stock removal cycle.

23 - 9xxx Progs Edit Lock

When this setting is **ON**, the control does not let you view or alter the files in the 09000 directory in **Memory**/. This protects macro programs, probing cycles, and any other files in the 09000 folder.

If you attempt to access the 09000 folder while Setting 23 is **ON**, you get the message *Setting 23 restricts access to folder.*

28 - Can Cycle Act w/o X/Y

This is an **ON/OFF** setting. The preferred setting is **ON**.

When it is **OFF**, the initial canned cycle definition block requires an *X* or *Y* code for the canned cycle to be executed.

When it is **ON**, the initial canned cycle definition block causes one cycle to be executed even when there is no *X* or *Y* code in the block.



NOTE:

*When an **L0** is in that block, it will not execute the canned cycle on the definition line. This setting has no effect on *G72* cycles.*

29 - G91 Non-modal

Turning this setting **ON** uses the G91 command only in the program block it is in (non-modal). When it is **OFF**, and a G91 is commanded, the machine uses incremental moves for all axis positions.



*This setting must be **OFF** for G47 engraving cycles.*

31 - Reset Program Pointer

When this setting is **OFF**, **[RESET]** does not change the position of the program pointer. When it is **ON**, pressing **[RESET]** moves the program pointer to the beginning of the program.

32 - Coolant Override

This setting controls how the coolant pump operates. When Setting 32 **NORMAL**, you can press **[COOLANT]**, or you can use M-codes in a program, to turn the coolant pump on and off.

When Setting 32 is **OFF**, the control gives the message *FUNCTION LOCKED* when you press **[COOLANT]**. The control gives an alarm when a program commands the coolant pump on or off.

When Setting 32 is **IGNORE**, the control ignores all programmed coolant commands, but you can press **[COOLANT]** to turn the coolant pump on or off.

39 - Beep @ M00, M01, M02, M30

Turning this setting **ON** causes the keyboard beeper to sound when an M00, M01 (with Optional Stop active), M02, or an M30 is found. The beeper continues until a button is pressed.

42 - M00 After Tool Change

Turning this setting **ON** stops the program after a tool change and a message is displayed stating this. **[CYCLE START]** must be pressed to continue the program.

43 - Cutter Comp Type

This controls how the first stroke of a compensated cut begins and the way the tool is cleared from the part. The selections can be **A** or **B**; see the Tool Nose Compensation section on page 137.

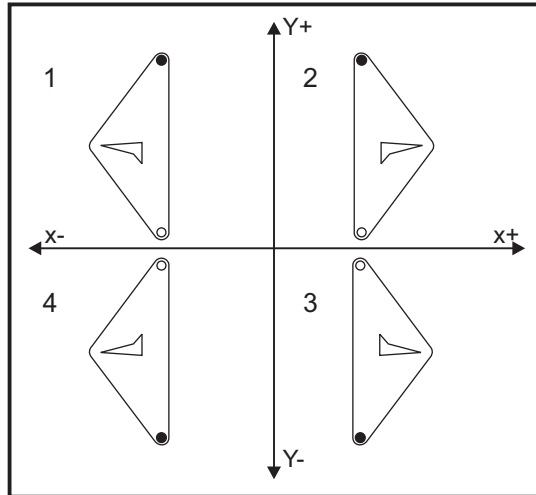
44 - Min F In Radius CC %

Minimum feedrate in radius tool nose compensation percent affects the feedrate when cutter compensation moves the tool towards the inside of a circular cut. This type of cut slows down to maintain a constant surface feedrate. This setting specifies the slowest feedrate as a percentage of the programmed feedrate.

45, 46, 47 - Mirror Image X, Y, Z Axis

When one or more of these settings is **ON**, axis motion is mirrored (reversed) around the work zero point. See also G101, Enable Mirror Image.

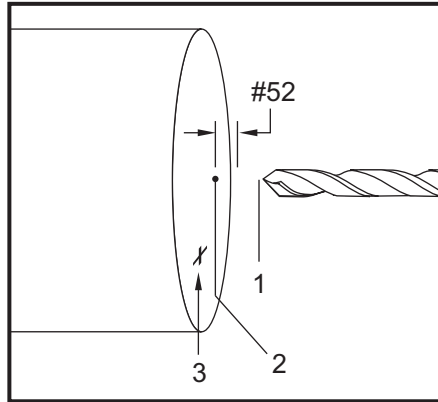
F9.2: No Mirror Image [1], Setting 45 **ON** - X Mirror [2], Setting 46 **ON** - Y Mirror [4], Setting 45 and Setting 46 **ON** - XY Mirror [3]



52 - G83 Retract Above R

This setting changes the way G83 (peck drilling cycle) behaves. Most programmers set the reference (R) plane well above the cut to ensure that the chip clearing motion actually allows the chips to get out of the hole. However this wastes time as the machine drills through empty distance. If Setting 52 is set to the distance required to clear chips, the R plane can be put closer to the part being drilled.

F9.3: Setting 52 - G83 Retract Above R: [#52] Setting 52, [1] Start position, [2] R plane, [3] Face of the part.



53 - Jog w/o Zero Return

Turning this setting **ON** allows the axes to be jogged without zero returning the machine (finding machine home). This is a dangerous condition as the axis can be run into the mechanical stops and possibly damage the machine. When the control is powered up, this setting automatically returns to **OFF**.

56 - M30 Restore Default G

When this setting is **ON**, ending a program with M30 or pressing **[RESET]** returns all modal G-codes to their defaults.

57 - Exact Stop Canned X-Z

The rapid XZ motion associated with a canned cycle may not achieve an exact stop when this setting is **OFF**. Turning this setting **ON** makes the XZ motion come to an exact stop.

58 - Cutter Compensation

This setting selects the type of cutter compensation used (FANUC or YASNAC). Refer to the tool functions section on page 133.

59, 60 - Probe Offset X+, X-

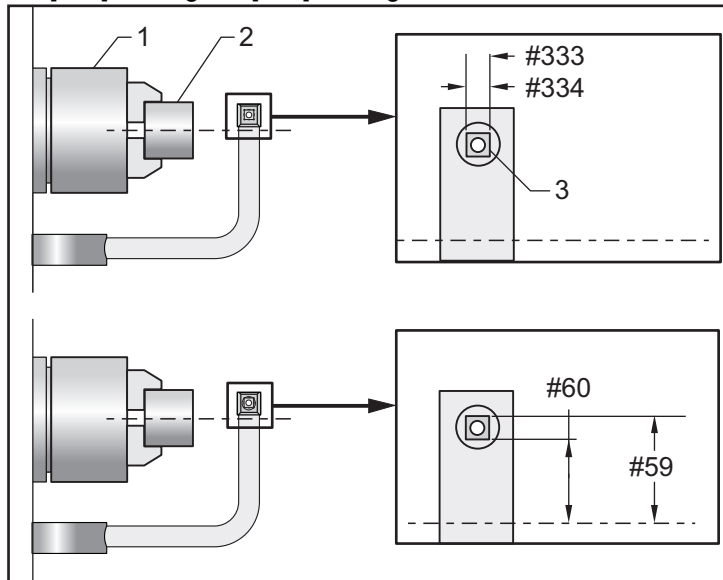
These settings are used to define the displacement and size of the ATP. These four settings (59, 60, 333, 334) specify the travel distance and direction from where the probe is triggered to where the actual sensed surface is located.

For more information on how to calibrate the ATP refer to page **179**.

These settings are used by the G31 code. The values entered for each setting must be positive numbers.

Macros can be used to access these settings, see the Macro section for more information.

F9.4: 59/60/X###/## Tool Probe Offset:[1] Chuck, [2] Part, [3] Probe, [#59] Setting 59, [#60] Setting 60, [###] Setting ##, [###] Setting ##,



63 - Tool Probe Width

This setting is used to specify the width of the probe used to test tool diameter. This setting only applies to the probing option.

For more information on how to calibrate the ATP refer to page **179**.

64 - T. Ofs Meas Uses Work

The (Tool Offset Measure Uses Work) setting changes the way the **[Z FACE MEASURE]** key works. When this is **ON**, the entered tool offset is the measured tool offset plus the work coordinate offset (Z-axis). When it is **OFF**, the tool offset equals the Z machine position.

74 - 9xxx Progs Trace

This setting, along with Setting 75, is useful for debugging CNC programs. When Setting 74 is **ON**, the control displays the code in the macro programs (O9xxxx). When the setting is **OFF**, the control does not display the 9000 series code.

75 - 9xxxx Progs Single BLK

When Setting 75 is **ON** and the control is operating in Single Block mode, then the control stops at each block of code in a macro program (O9xxxx) and waits for the operator to press **[CYCLE START]**. When Setting 75 is **OFF** the macro program is run continuously, the control does not pause at each block, even if Single Block is **ON**. The default setting is **ON**.

When Setting 74 and Setting 75 are both **ON**, the control acts normally. That is, all blocks executed are highlighted and displayed, and when in Single-Block mode there is a pause before each block is executed.

When Setting 74 and Setting 75 are both **OFF**, the control executes 9000 series programs without displaying the program code. If the control is in Single-Block mode, no single-block pause occurs during the running of the 9000 series program.

When Setting 75 is **ON** and Setting 74 is **OFF**, 9000 series programs are displayed as they are executed.

77 - Scale Integer F

This setting allows the operator to select how the control interprets an **F** value (feedrate) that does not contain a decimal point. (It is recommended that you always use a decimal point.) This setting helps operators run programs developed on a control other than Haas.

There are 5 feedrate settings. This chart shows the effect of each setting on a given F10 address.

INCH		MILLIMETER	
Setting 77	Feedrate	Setting 77	Feedrate
DEFAULT	F0.0010	DEFAULT	F0.0100
INTEGER	F10.	INTEGER	F10.
.1	F1.0	.1	F1.0
.01	F0.10	.01	F0.10

INCH		MILLIMETER	
.001	F0.010	.001	F0.010
.0001	F0.0010	.0001	F0.0010

80 - Mirror Image B Axis

This is an **ON/OFF** setting. When it is **OFF**, axis motions occur normally. When it is **ON**, B-Axis motion may be mirrored (or reversed) around the work zero point. Also, see G101 and Settings 45, 46, 47, 48, and 250.

82 - Language

Languages other than English are available in the Haas control. To change to another language, choose a language with the **[LEFT]** and **[RIGHT]** cursor arrows, then press **[ENTER]**.

83 - M30/Resets Overrides

When this setting is **ON**, an **M30** restores any overrides (feedrate, spindle, rapid) to their default values (100%).

84 - Tool Overload Action

When a tool becomes overloaded, Setting 84 designates the control response. These settings cause specified actions (refer to the Advanced Tool Management Introduction

on page **102**):

- **ALARM** causes the machine to stop.
- **FEEDHOLD** displays the message *Tool Overload* and the machine stops in a feedhold situation. Press any key to clear the message.
- **BEEP** causes an audible noise (beep) from the control.
- **AUTOFEED** causes the control to automatically limit the feedrate based on the tool load.



NOTE:

*When tapping (rigid or floating), the feed and spindle overrides are locked out, so the **AUTOFEED** setting is ineffective (the control appears to respond to the override buttons, by displaying the override messages).*

**CAUTION:**

*Do not use the **AUTOFEED** setting when thread milling or auto reversing tapping heads, as it can cause unpredictable results or even a crash.*

The last commanded feedrate is restored at the end of program execution, or when the operator presses **[RESET]** or turns **OFF** the **AUTOFEED** setting. The operator can use **[FEEDRATE OVERRIDE]** while the **AUTOFEED** setting is selected. These keys are recognized by the **AUTOFEED** setting as the new commanded feedrate as long as the tool load limit is not exceeded. However, if the tool load limit has already been exceeded, the control ignores **[FEEDRATE OVERRIDE]**.

85 - Maximum Corner Rounding

This setting defines the machining accuracy tolerance around corners. The initial default value is 0.05". This means that the control keeps the radii of corners no bigger than 0.05".

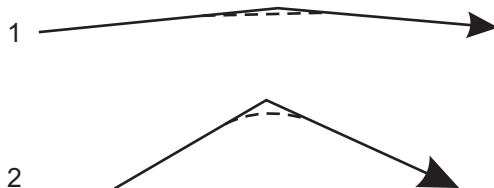
Setting 85 causes the control to adjust feeds around corners to meet the tolerance value. The lower the value of Setting 85, the slower the control feeds around corners to meet the tolerance. The higher the value of Setting 85, the faster the control feeds around corners, up to the commanded feedrate, but it could round the corner off to a radius up to the tolerance value.

**NOTE:**

The angle of the corner also affects the change to the feedrate. The control can cut shallow corners within tolerance at a higher feedrate than it can with tighter corners.

F9.5:

The control can cut corner [1] within tolerance at a higher feedrate than it can cut corner [2].

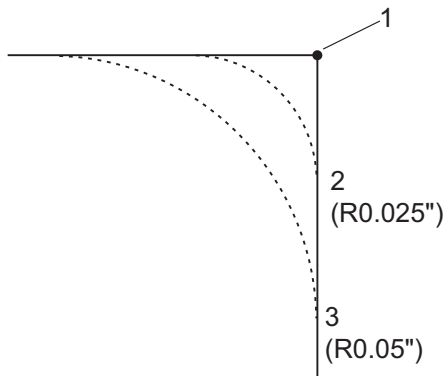


If Setting 85 has a value of zero, the control acts as if exact stop is active in each motion block.

Refer also to G187 - Accuracy Control (Group 00) on page 324.

F9.6:

Assume that the commanded feedrate is too high to achieve corner [1]. If Setting 85 has a value of 0.025, then the control slows the feedrate enough to achieve corner [2] (with a radius of 0.025"). If Setting 85 has a value of 0.05, then the control slows the feedrate enough to achieve corner [3]. The feedrate to achieve corner [3] is faster than the feedrate to achieve corner [2].



87 - Tool Change Resets Override

This is an **ON/OFF** setting. When a **Tnn** tool change is executed and this setting is **ON**, any overrides are canceled and set to their programmed values.

**NOTE:**

This setting only affects programmed tool changes, it does not affect [TURRET FWD] or [TURRET REV] tool changes.

88 - Reset Resets Overrides

This is an **ON/OFF** setting. When it is **ON** and **[RESET]** is pressed, any overrides are canceled and set to their programmed values or defaults (100%).

90 - Max Tools To Display

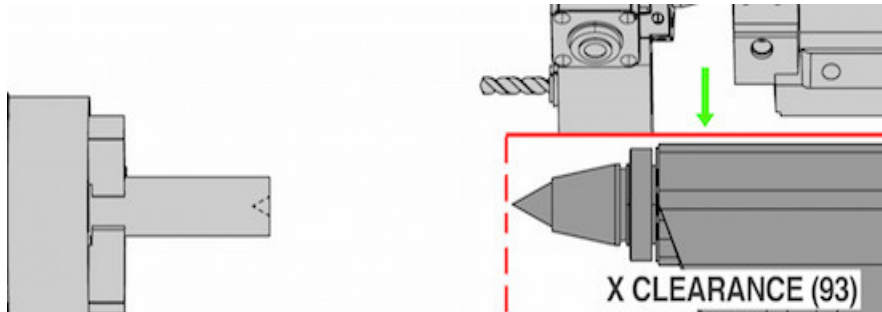
This setting limits the number of tools displayed on the Tool Offsets screen.

93 - Tailstock X Clearance

This setting works with Setting 94 to define a tailstock travel restriction zone that limits interaction between the tailstock and the tool turret. This setting determines the X-axis travel limit when the difference between the Z-axis location and the tailstock location falls below the value in Setting 94. If this condition occurs and a program is running then an alarm is generated. When jogging, no alarm is generated, but travel will be limited.

F9.7:

Tailstock X Clearance

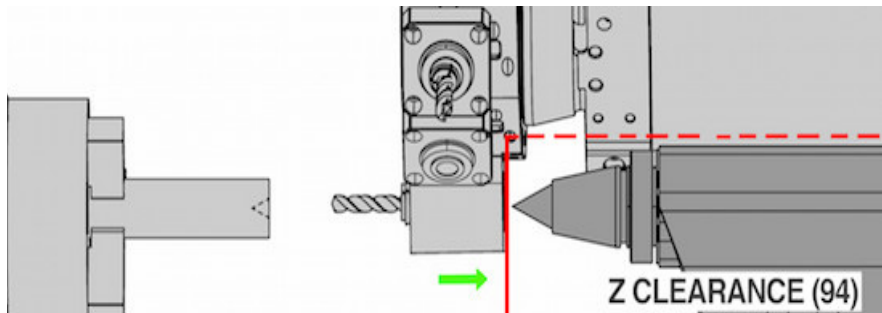


94 - Tailstock Z Clearance

This setting is the minimum allowable difference between the Z-axis and the tailstock (see Setting 93). If units are in inches, a value of -1.0000 means that when the X-axis is below the X clearance plane (Setting 93), the Z-axis must be more than 1 inch away from the tailstock position in the Z-axis negative direction.

F9.8:

Tailstock Z Clearance



95 - Thread Chamfer Size

This setting is used in G76 and G92 threading cycles when an M23 is commanded. When command M23 is active, threading strokes end with an angled retraction, as opposed to pulling straight out. The value in Setting 95 is equal to the number of turns (chamfered threads) desired.

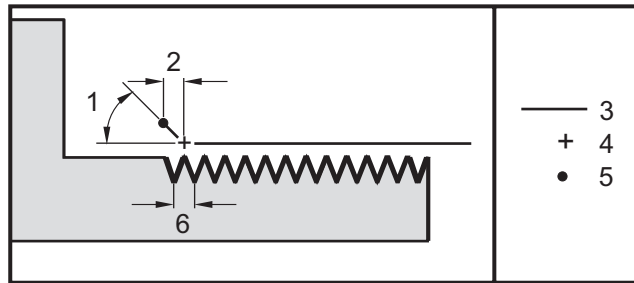


NOTE:

Settings 95 and 96 interact with each other. (Multiple of current thread lead, F or E).

F9.9:

Setting 95 - Thread Chamfer Size, G76 or G92 threading stroke with M23 active: [1] Setting 96 = 45, [2] Setting 95 x Lead, [3] Tool path, [4] Programmed thread endpoint, [5] Actual stroke endpoint, [6] Lead.



96 - Thread Chamfer Angle

See Setting 95.

97 - Tool Change Direction

This setting determines the default tool change direction. It may be set to either **SHORTEST** or M17/M18.

When **SHORTEST** is selected, the control turns in the direction necessary to reach the next tool with the least movement. The program can still use M17 and M18 to fix the tool change direction, but once this is done it is not possible to revert back to the shortest tool direction other than **[RESET]** or M30/M02.

Selecting M17/M18, the control moves the tool turret either always forward or always reverse based on the most recent M17 or M18. When **[RESET]**, **[POWER ON]**, or M30/M02 is executed, the control assumes M17 as the tool turret direction during tool changes, always forward. This option is useful when a program must avoid certain areas of the tool turret due to odd-sized tools.